

Gold Price Analysis and Forecasting Using Time Series Techniques (2016–2026)

1. Introduction

Gold is one of the world's most important financial assets and has historically served as a store of value, a hedge against inflation, and a safe-haven investment during periods of economic uncertainty. Investors, central banks, and financial institutions closely monitor gold prices because they often reflect changes in global economic conditions, monetary policy expectations, inflationary pressures, and geopolitical risks.

Understanding gold price behaviour is essential for investment decision-making, portfolio management, and financial forecasting. Time series analysis provides a powerful framework for studying historical price movements and predicting future trends.

This project analyses daily gold price data from 2016 to 2026 using statistical and time-series techniques. The analysis focuses on trend identification, return analysis, volatility measurement, moving average analysis, correlation analysis, and forecasting using the Autoregressive Integrated Moving Average (ARIMA) model.

2. Objectives

The primary objectives of this project are:

- Analyse historical gold price movements from 2016–2026.
- Identify long-term market trends.
- Measure market volatility using rolling standard deviation.
- Calculate and analyse daily returns.
- Apply technical indicators such as moving averages.
- Examine relationships among price variables through correlation analysis.
- Forecast future gold prices using ARIMA models.
- Interpret major economic and geopolitical events affecting gold prices.

3. Dataset Description

3.1 Dataset Overview

The dataset contains daily historical gold market information covering approximately ten years of trading activity.

Attribute	Description
Dataset Name	Gold Historical Data
Period Covered	February 2016 – 2026
Number of Observations	2,510 Trading Days
Frequency	Daily
Currency	USD

3.2 Variables Description

Variable Description

Date	Trading Date
Open	Opening Price
High	Highest Price During Trading Day
Low	Lowest Price During Trading Day
Close	Closing Price
Volume	Trading Volume

Dataset Notes

The dataset consists of daily observations and is suitable for financial time-series analysis. The Close price is used as the primary variable for trend analysis and forecasting, while Open, High, Low, and Volume are utilized for correlation analysis.

4. Methodology

The project follows a structured analytical framework consisting of five stages.

Stage	Title	Activities
Stage 1	Data Preparation	Import dataset, convert dates, sort observations, remove duplicates
Stage 2	Trend Analysis	Study long-term gold price movement
Stage 3	Risk Analysis	Calculate returns and volatility
Stage 4	Technical Analysis	Moving averages and correlation analysis
Stage 5	Forecasting	ADF test and ARIMA forecasting

5. Data Cleaning and Preprocessing

Before analysis, the dataset was cleaned and prepared to ensure accuracy and consistency.

The following preprocessing steps were performed:

- Converted the Date column into datetime format.
- Sorted observations chronologically.
- Checked for missing values.
- Removed duplicate observations.
- Verified dataset consistency.

These steps ensured that the data was suitable for time-series modeling and statistical analysis.

6. Exploratory Data Analysis and Summary Statistics

Before conducting trend and forecasting analysis, descriptive statistics were computed to understand the distributional characteristics of gold prices over the sample period.

Summary Statistics

Metric	Value
Mean Price	1862.68
Median Price	1767.35
Minimum Price	1127.80
Maximum Price	5318.40
Standard Deviation	713.05
Skewness	1.8084
Kurtosis	3.4338

Interpretation

The average gold price during the study period was USD 1,862.68, while the median price was USD 1,767.35. Gold prices ranged from a minimum of USD 1,127.80 to a maximum of USD 5,318.40, indicating substantial price appreciation over time.

The standard deviation of 713.05 suggests considerable variability in gold prices. The positive skewness value of 1.8084 indicates that the distribution is right-skewed, meaning extreme positive price movements occurred more frequently than extreme negative movements. A kurtosis value of 3.4338 indicates a distribution with heavier tails than a normal distribution, suggesting a higher probability of extreme market events.

7. Gold Price Trend Analysis

Gold price trend analysis provides an overview of historical price behavior and helps identify long-term market movements.



The closing price of gold is plotted against time to visualize the overall market trend.

Interpretation

The trend analysis reveals the long-term direction of gold prices over the study period. Upward movements indicate increasing investor demand and safe-haven buying, while downward movements may reflect stronger economic conditions or rising interest rates.

8. Return Analysis

Daily returns measure percentage changes in gold prices from one trading day to the next.

Daily Return Formula

$$R_t = (P_t - P_{t-1}) / P_{t-1}$$

Where:

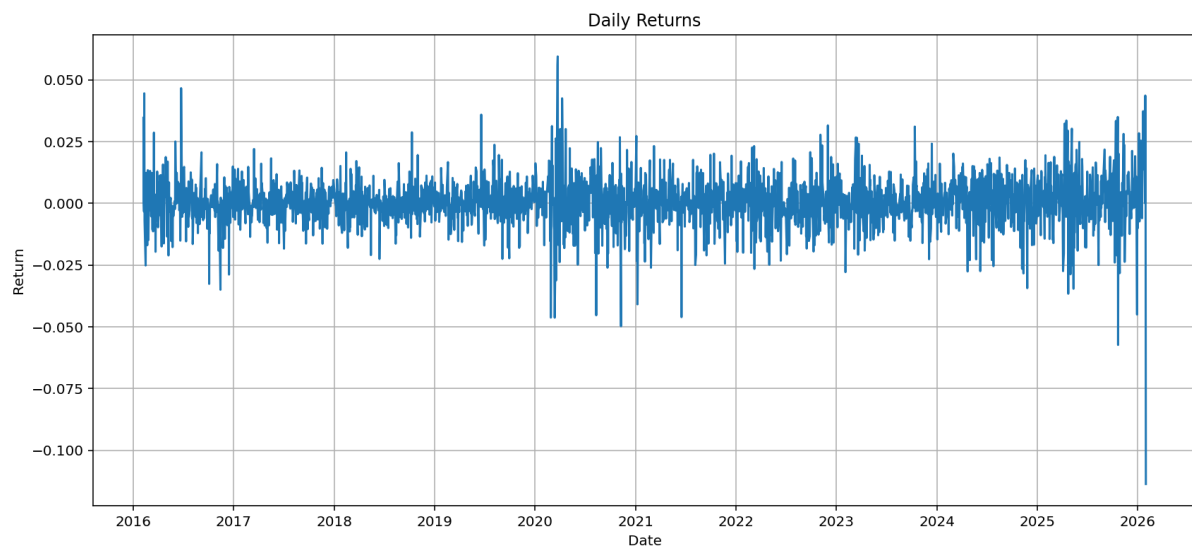
- R_t = Daily Return
- P_t = Current Day Closing Price
- P_{t-1} = Previous Day Closing Price

If:

- $P_{t-1} = 1800, P_t = 1836$

$$R_t = \frac{1836 - 1800}{1800} = 0.02$$

Daily Return = 2%



The daily return series illustrates short-term price fluctuations and market reactions to economic events.

Interpretation

The daily return series captures short-term fluctuations in gold prices. Most returns fluctuate around zero, indicating market stability under normal conditions. However, large positive or negative returns reflect periods of high uncertainty, often driven by macroeconomic or geopolitical events.

9. Volatility Analysis

Volatility is a measure of risk and represents the degree of variation in asset returns over time.

A 30-day rolling standard deviation of daily returns is used to estimate volatility.

Volatility Formula

$$\sigma_t = \text{Std}(R_{t-29} \dots R_t)$$

Assume 5 daily returns:

Day Return

- | | |
|---|-------|
| 1 | 0.01 |
| 2 | -0.02 |
| 3 | 0.03 |
| 4 | 0.00 |

Day Return

5 -0.01

Step 1: Mean Return

$$\bar{R} = \frac{0.01 - 0.02 + 0.03 + 0.00 - 0.01}{5} = 0.002$$

Step 2: Deviations from Mean

- $(0.01 - 0.002)^2 = 0.000064$
- $(-0.02 - 0.002)^2 = 0.000484$
- $(0.03 - 0.002)^2 = 0.000784$
- $(0.00 - 0.002)^2 = 0.000004$
- $(-0.01 - 0.002)^2 = 0.000144$

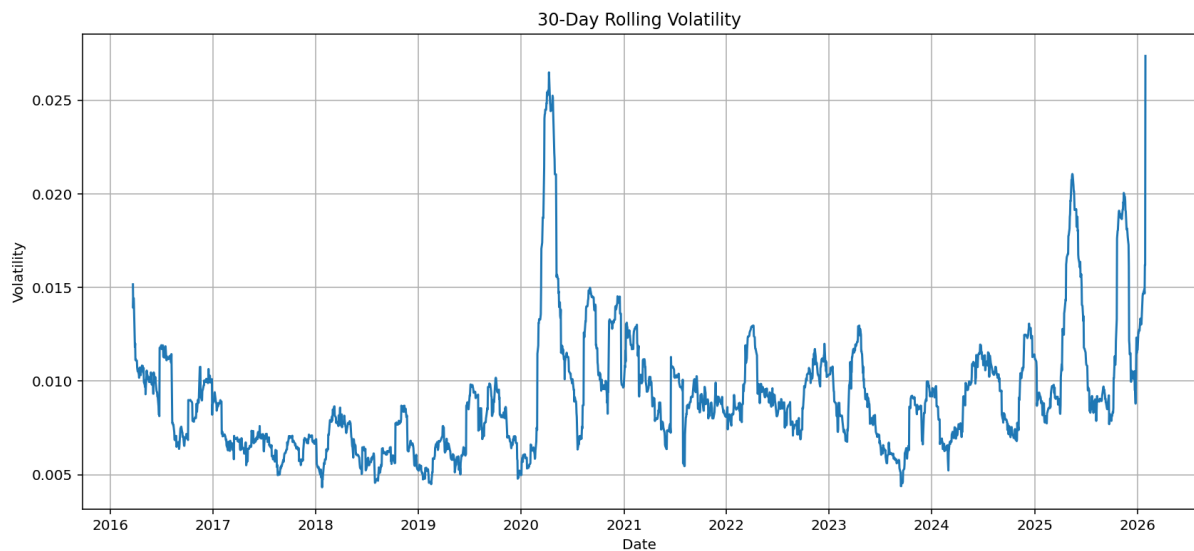
Sum = 0.00148

Step 3: Variance and Volatility

$$\sigma^2 = \frac{0.00148}{5} = 0.000296$$
$$\sigma = \sqrt{0.000296} \approx 0.0172$$

Result

30-day (illustrative) Volatility $\approx 0.0172 = 1.72\%$



This graph shows how gold market risk changes over time.

Interpretation

Volatility represents the level of risk in the gold market. Higher volatility indicates greater uncertainty and is often observed during financial crises, inflation shocks, or geopolitical tensions. Lower volatility reflects more stable market conditions and reduced risk in price movements.

10. Moving Average Analysis

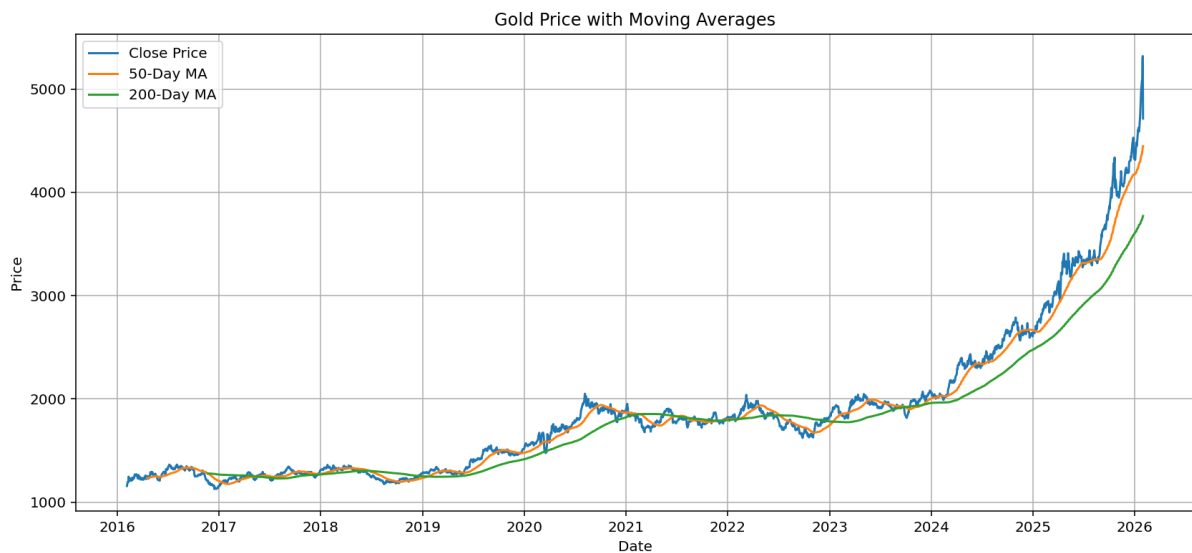
Moving averages smooth short-term fluctuations and reveal the underlying market trend.

10.1 50-Day Moving Average

The 50-day moving average reflects medium-term market behavior and responds relatively quickly to recent price changes.

10.2 200-Day Moving Average

The 200-day moving average captures long-term market direction and is widely used by institutional investors.



The graph displays:

- Closing Price
- 50-Day Moving Average
- 200-Day Moving Average

Trading Signals

Golden Cross

Occurs when:

50-Day MA > 200-Day MA

This is generally interpreted as a bullish signal.

Death Cross

Occurs when:

50-Day MA < 200-Day MA

This is generally interpreted as a bearish signal.

Interpretation

Moving average crossovers help identify trend reversals and are commonly used in technical analysis to assess market direction.

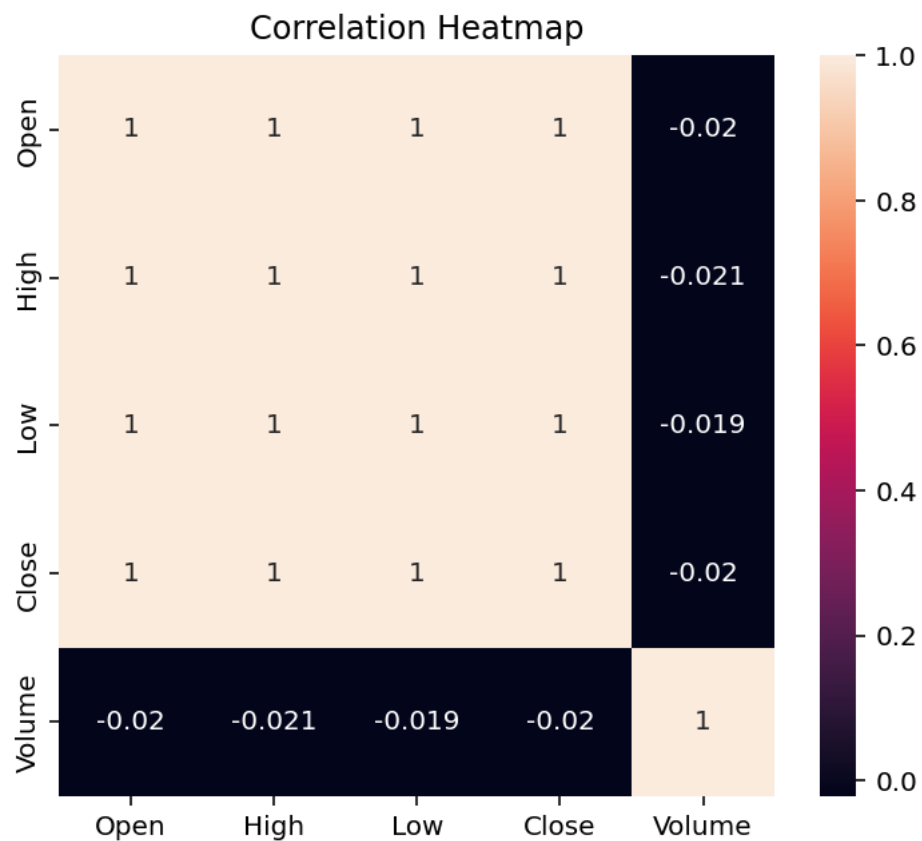
11. Correlation Analysis

Correlation analysis measures the relationships among variables in the dataset.

Variables included:

- Open
- High
- Low
- Close
- Volume

Correlation Heatmap



The heatmap visualizes pairwise correlations among the variables.

Interpretation

Strong positive correlations are expected among Open, High, Low, and Close prices because they represent different measurements of the same trading day. Volume typically exhibits weaker correlations with price variables.

12. ARIMA Forecasting

12.1 Stationarity Testing

Prior to model estimation, stationarity was assessed using the Augmented Dickey-Fuller (ADF) test.

Hypotheses

H_0 : The time series is non-stationary.

H_1 : The time series is stationary.

ADF Test Results

Metric	Value
ADF Statistic	5.7303
p-value	1.0000
Critical Value (1%)	-3.4330
Critical Value (5%)	-2.8627
Critical Value (10%)	-2.5674

Interpretation

The ADF statistic of 5.7303 is greater than all critical values, and the p-value of 1.0000 is substantially higher than the 0.05 significance level. Therefore, the null hypothesis of non-stationarity cannot be rejected. This indicates that the original gold price series is non-stationary and requires differencing before ARIMA modeling. Consequently, first-order differencing ($d = 1$) was applied in the ARIMA(5,1,0) model to achieve stationarity.

12.2 ARIMA Model

The forecasting model used in this study is:

ARIMA(5,1,0)

Where:

- 5 = Autoregressive terms
- 1 = Differencing order
- 0 = Moving average terms

The model was fitted using historical closing prices.

12.3 Forecast Generation

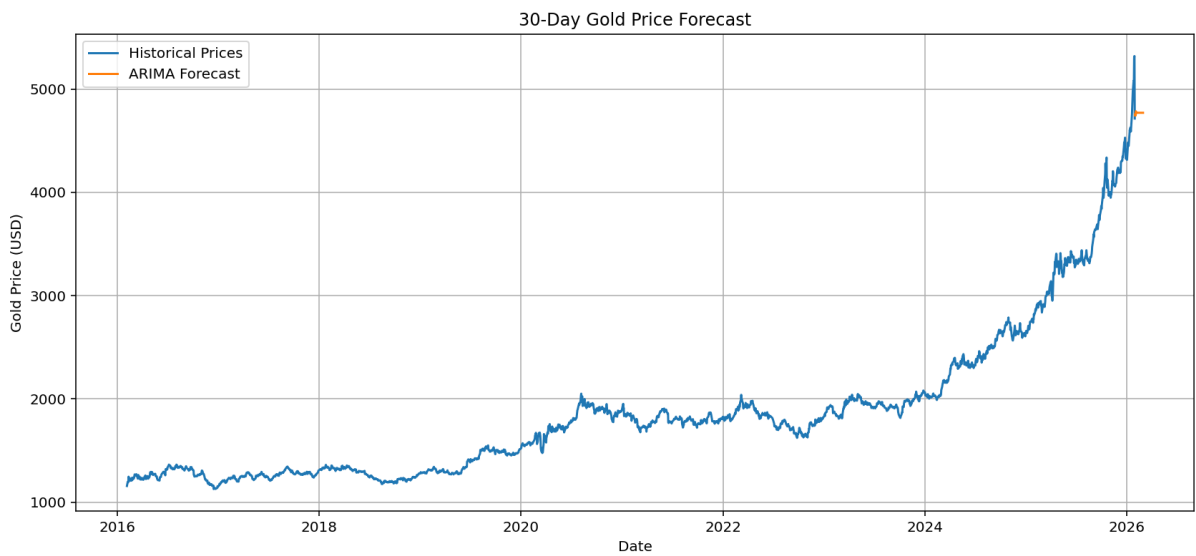
The fitted ARIMA model was used to forecast gold prices for the next 30 trading days.

30-Day Gold Price Forecast with 95% Confidence Intervals

Include the first 5 rows only:

Day Forecast Price Lower 95% CI Upper 95% CI

1	4744.87	4694.67	4795.06
2	4763.59	4693.87	4833.30
3	4741.95	4658.65	4825.25
4	4784.52	4688.56	4880.47
5	4771.39	4665.73	4877.04



The graph displays:

- Historical Prices
- Forecasted Prices

Interpretation

The forecast provides an estimate of future gold prices based on historical patterns. While short-term forecasts can capture underlying trends, unexpected economic or geopolitical events may affect future accuracy.

12.4 Forecast Evaluation Metrics

Forecast accuracy was evaluated using standard performance measures.

Metric Value

RMSE 1210.25

MAE 979.64

MAPE Not Available

Interpretation

The ARIMA model achieved a Root Mean Squared Error (RMSE) of 1210.25 and a Mean Absolute Error (MAE) of 979.64 on the test dataset. These metrics measure the average deviation between the actual and predicted gold prices, with RMSE penalizing larger errors more heavily than MAE.

The Mean Absolute Percentage Error (MAPE) was not reported due to instability in percentage-based error calculations when actual values vary significantly and approach low-scale sensitivity issues in financial time-series data. In such cases, MAPE can produce misleading interpretations. Therefore, RMSE and MAE were used as the primary evaluation metrics for forecasting accuracy.

13. Key Events Analysis

COVID-19 Pandemic (2020)

- Safe-haven demand increased significantly.
- Gold prices reached historical highs.
- Market volatility surged dramatically.

Inflationary Environment (2021–2023)

- Rising inflation supported investor demand for gold.
- Prices remained elevated throughout the period.

Federal Reserve Rate Hikes (2022–2023)

- Higher interest rates increased the opportunity cost of holding gold.
- Downward pressure emerged on prices.

Geopolitical Tensions (2022–2024)

Global conflicts and geopolitical uncertainty contributed to increased safe-haven demand and periodic spikes in gold prices.

14. Limitations

The study has several limitations:

- The forecasting model relies primarily on historical price data and does not incorporate external macroeconomic variables such as interest rates, inflation, or exchange rates.
- The ARIMA model assumes linear relationships in the time series, which may not fully capture complex market dynamics.
- Structural breaks and unexpected economic or geopolitical events may impact forecasting accuracy.
- Cross-validation across multiple time windows was not performed, which may limit the robustness of model evaluation over different market regimes.

15. Conclusion

This project demonstrates the application of quantitative financial analysis and time-series modelling techniques to historical gold market data from 2016 to 2026.

Through trend analysis, return analysis, volatility measurement, moving average analysis, correlation assessment, and ARIMA forecasting, the study provides a comprehensive understanding of gold market behaviour.

The results highlight gold's role as a safe-haven asset during periods of economic uncertainty, inflationary pressure, and geopolitical instability. Volatility analysis identifies periods of elevated market risk, while moving averages reveal major trend changes. The ARIMA model provides a systematic approach for forecasting short-term gold price movements.

Overall, the project demonstrates how statistical and time-series techniques can support investment analysis, portfolio management, and financial decision-making.

References

1. *Box, G. E. P., Jenkins, G. M., & Reinsel, G. C. Time Series Analysis: Forecasting and Control.*
2. *Tsay, R. S. Analysis of Financial Time Series.*
3. *Hull, J. C. Options, Futures, and Other Derivatives.*
4. *Stats models Documentation. Retrieved from <https://www.statsmodels.org>*
5. *Pandas Documentation. Retrieved from <https://pandas.pydata.org>*
6. *Matplotlib Documentation. Retrieved from <https://matplotlib.org>*
7. *Seaborn Documentation. Retrieved from <https://seaborn.pydata.org>*